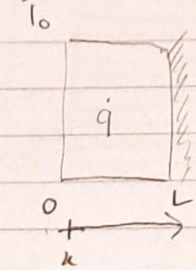
HW5:

3.77:



$$\dot{q} = \dot{q}_0\left(1 - \frac{x}{L}\right)$$

Balance eqn:

$$k\frac{d^2T}{dx^2} + \dot{q} = 0$$

B.C: $T(x=0) = T_0$

$$k\frac{dT}{dx}\bigg|_{x=L} = 0$$

$$k\frac{d^2T}{dx^2} + \dot{q}_0\left(1 - \frac{x}{L}\right) = 0$$

$$\frac{d^2T}{dx^2} = \frac{1}{kL}x - \frac{\dot{q}_0}{k}$$

$$\frac{dT}{dx} = \frac{1}{2kL}x^2 - \frac{\dot{q}_0}{k}x + A$$

$$T = \frac{1}{6kL}x^3 - \frac{\dot{q}_0}{2k}x^2 + Ax + B$$

B.C: $T(x=0) = T_0 \Rightarrow T_0 = B$

B.C: $k\frac{dT}{dx}\Big|_{x=L} = 0 \Rightarrow \frac{L^3}{6kL} - \frac{\dot{q}_0L^2}{2k} + AL + T_0 = 0$

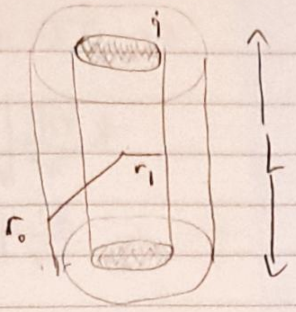
$$AL = -\frac{L^2}{6k} + \frac{\dot{q}_0L^2}{2k} - T_0 \Rightarrow A = \frac{-L}{6k} + \frac{\dot{q}_0L}{2k} - \frac{T_0}{L}$$

$$\Rightarrow T(x) = \frac{1}{6kL}x^3 - \frac{\dot{q}_0}{9k}x^2 + \left(\frac{-L}{6k} + \frac{\dot{q}_0L}{2k} - \frac{T_0}{L}\right)x + T_0$$

3.80:

T_∞

h



Balance eqn: $\frac{1}{r}\frac{d}{dr}\left(k r \frac{dT}{dr}\right) + \dot q = 0$

$$B.C: k\frac{dT}{dr}\Big|_{r=r_i} = 0$$

$$-k\frac{dT}{dr}\Big|_{r=r_o} = h(T(r=r_o) - T_\infty)$$

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) = \frac{-\dot q}{k} r$$

$$r\frac{dT}{dr} = \frac{-\dot q}{2k}r^2 + A \Rightarrow \frac{dT}{dr} = \frac{-\dot q}{2k}r + \frac{A}{r}$$

$$T = \frac{-\dot q}{4k}r^2 + A\ln r + B$$

$$B.C: k\frac{dT}{dr}\Big|_{r=r_i} = 0 \Rightarrow k\left(\frac{-\dot q}{2k}r_i + \frac{A}{r_i}\right) = 0 \Rightarrow \frac{-\dot q}{2k}r_i + \frac{A}{r_i} = 0$$

$$\frac{A}{r_i} = \frac{-\dot q}{2k}r_i \Rightarrow A = \frac{-\dot q}{2k}r_i^2$$

$$\Rightarrow T = \frac{-\dot q}{4k}r^2 - \frac{\dot q r_i^2}{2k}\ln r + B$$

$$B.C: -k\frac{dT}{dr}\Big|_{r=r_o} = h(T(r=r_o) - T_\infty)$$

$$-k\left(\frac{-\dot q}{2k}r_o\right) = h\left(\frac{-\dot q r_o^2}{4k} - \frac{\dot q r_i^2}{2k}\ln r_o + B - T_\infty\right)$$

$$\Rightarrow \frac{-\dot q r_o}{2h} = \frac{-\dot q r_o^2}{4k} - \frac{\dot q r_i^2 \ln r_o}{2k} + B - T_\infty$$

$$\Rightarrow B = \frac{-\dot q r_o}{2h} + \frac{\dot q r_o^2}{4k} + \frac{\dot q r_i^2 \ln r_o}{2k} + T_\infty$$

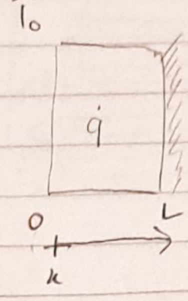
$$\Rightarrow T(r) = \frac{-\dot q}{4k}r^2 - \frac{\dot q r_i^2}{2k}\ln r + \frac{\dot q r_o^2}{4k} - \frac{\dot q r_o}{2h} + \frac{\dot q r_i^2 \ln r_o}{2k} + T_\infty$$

b) heatrate $q'(r_o)$: $-k\frac{dT}{dr}\Big|_{r=r_o}(S.A)$

$$= -k\left(\frac{-\dot q r_o}{2k} - \frac{\dot q r_i^2}{2k}\frac{1}{r_o}\right)(2\pi r_o L) = (\dot q r_o^2 + \dot q r_i^2)\pi L$$

HW 5:

3.77:



$$\dot{q} = \dot{q}_0\left(1 - \frac{x}{L}\right)$$

Balance eqn:

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$$T = \frac{1}{6kL}x^3 - \frac{\dot{q}_0}{2k}x^2 + Ax + B$$

B.C: $T(x=0) = T_0 \Rightarrow T_0 = B$

B.C: $k\frac{dT}{dx}\Big|_{x=L} = 0 \Rightarrow \frac{L^3}{6kL} - \frac{\dot{q}_0 L^2}{2k} + AL + T_0 = 0$

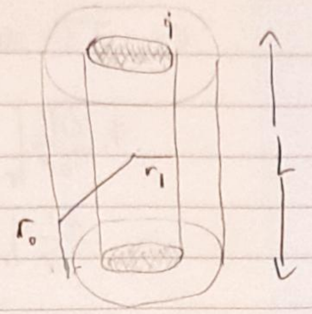
$$AL = -\frac{L^2}{6k} + \frac{\dot{q}_0 L^2}{2k} - T_0 \Rightarrow A = \frac{-L}{6k} + \frac{\dot{q}_0 L}{2k} - \frac{T_0}{L}$$

$$\Rightarrow T(x) = \frac{1}{6kL}x^3 - \frac{\dot{q}_0}{2k}x^2 + \left(\frac{-L}{6k} + \frac{\dot{q}_0 L}{2k} - \frac{T_0}{L}\right)x + T_0$$

3.80:

T_∞

h



Balance eqn: $\frac{1}{r}\frac{d}{dr}\left(k r \frac{dT}{dr}\right) + \dot{q} = 0$

B.C: $k\frac{dT}{dr}\Big|_{r=r_i} = 0$

$-k\frac{dT}{dr}\Big|_{r=r_o} = h(T(r=r_o) - T_\infty)$

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) = \frac{-\dot{q}}{k} r$$

$$r\frac{dT}{dr} = \frac{-\dot{q}}{2k}r^2 + A \Rightarrow \frac{dT}{dr} = \frac{-\dot{q}}{2k}r + \frac{A}{r}$$

$$T = \frac{-\dot{q}}{4k}r^2 + A\ln r + B$$

B.C: $k\frac{dT}{dr}\Big|_{r=r_i} = 0 \Rightarrow k\left(\frac{-\dot{q}}{2k}r_i + \frac{A}{r_i}\right) = 0 \Rightarrow \frac{-\dot{q}}{2k}r_i + \frac{A}{r_i} = 0$

$$\frac{A}{r_i} = \frac{\dot{q}}{2k}r_i \Rightarrow A = \frac{-\dot{q}}{2k}r_i^2$$

$$\Rightarrow T = \frac{-\dot{q}}{4k}r^2 - \frac{\dot{q}r_i^2}{2k}\ln r + B$$

B.C: $-k\frac{dT}{dr}\Big|_{r=r_o} = h(T(r=r_o) - T_\infty)$

$$-k\left(\frac{-\dot{q}}{2k}r_o\right) = h\left(\frac{-\dot{q}r_o^2}{4k} - \frac{\dot{q}r_i^2}{2k}\ln r_o + B - T_\infty\right)$$

$$\Rightarrow \frac{-\dot{q}r_o}{2h} = \frac{-\dot{q}r_o^2}{4k} - \frac{\dot{q}r_i^2\ln r_o}{2k} + B - T_\infty$$

$$\Rightarrow B = \frac{-\dot{q}r_o}{2h} + \frac{\dot{q}r_o^2}{4k} + \frac{\dot{q}r_i^2\ln r_o}{2k} + T_\infty$$

$$\Rightarrow T(r) = \frac{-\dot{q}}{4k}r^2 - \frac{\dot{q}r_i^2}{2k}\ln r + \frac{\dot{q}r_o^2}{4k} - \frac{\dot{q}r_o}{2h} + \frac{\dot{q}r_i^2\ln r_o}{2k} + T_\infty$$

b) heatrate $q'(r_o) = -k\frac{dT}{dr}\Big|_{r=r_o}$ (S.A)

$$= -k\left(\frac{-\dot{q}r_o}{2k} - \frac{\dot{q}r_i^2}{2k}\frac{1}{r_o}\right)(2\pi r_o L) = (\dot{q}r_o^2 + \dot{q}r_i^2)\pi L$$